

# $\mathcal{L}_1$ Adaptive Control-Based Formation Tracking of Multiple Quadrotors Without Linear Velocity Feedback Under Unknown Disturbances

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**Abstract**—This paper addresses the problem of formation control for a quadrotor swarm (QS) system with directed graph topology under external environmental disturbances and unreliable internal state acquisition. The proposed distributed robust control framework, based on a geometric controller, incorporates  $\mathcal{L}_1$  adaptive controllers and differentiator systems. First, the geometric formation controller is designed to implement the formation control of the nominal system. Then,  $\mathcal{L}_1$  adaptive controllers are designed separately for each quadrotor's position loop and attitude loop subsystems to address the effects of uncertainties such as external time-varying disturbances (matched and unmatched disturbances) and different mass variations of quadrotors. Furthermore, the differentiator system is devised to accurately estimate the higher-order derivatives of the non-directly-measurable velocity information and the virtual translation control signal, which enhances system accuracy while reducing computational complexity. The Lyapunov stability theory is employed to analyze the stability of the closed-loop system. Finally, the effectiveness and exceptional performance of this approach in QS formation control were validated through numerical simulation and experimental results.

**Note to Practitioners**—The inspiration for this article comes from the issue of formation control in a cluster of quadrotor drones, which is also applicable to formation control in other types of drones. In this paper, a formation control algorithm based on  $\mathcal{L}_1$  adaptive control strategy and arbitrary-order differentiation is designed. This algorithm can address not only the issue of time-varying wind disturbances frequently

encountered during quadrotor drone flights but also the effects of unpredictable velocities and inconsistent masses of quadrotor drones. The disturbance rejection capability of this scheme enables quadrotor drones to be applied more safely and reliably in complex environments for search and rescue missions and surveillance tasks. Eliminating the need for linear velocity measurements reduces sensor costs and enhances system reliability and stability. The proposed formation control scheme allows the QS system to have different masses for each UAV, which can be applied to tasks such as collaboration logistics transportation, material delivery and crop spraying. Preliminary physical experiments have validated the feasibility of the proposed scheme, although it has not been applied in practical scenarios yet. In future research, we intend to equip each drone in the QS system with objects of different masses to achieve collaboration material transportation and delivery in complex environments.

**Index Terms**—Quadrotor formation control,  $\mathcal{L}_1$  adaptive, external disturbances, differentiator, directed topology.

## I. INTRODUCTION

**D**UE to the quadrotor's advantages of small size, lightweight, and adaptability to vertical take-off and landing, etc. The quadrotor has found extensive applications in both military and civilian domains, encompassing reconnaissance and surveillance operations, ground attacks, among others [1], [2]. Compared with the single UAV platform, the UAV swarm system, consisting of multiple quadrotors, offers powerful autonomy, versatile functions, and robust resistance to destruction, which can perform a more challenging and broader range of tasks [3]. Formation control is one of the critical technologies for generating, maintaining, and reconfiguring formation for UAV swarm systems. Virtual leader-follower is a better formation control strategy for accomplishing complex formation tasks due to its simple implementation, and flexible adjustment according to actual needs [4], [5], [6], [7], [8]. The directed graph in the formation control topology not only saves communication resources but also enhances flexibility and robustness, making it more suitable for quadrotor formation control with limited onboard resources [9], [10], [11]. However, formation control still faces the dual challenges of external complex environmental disturbances and unreliable internal state estimation that affects the safe flight of QS systems. Therefore, how to overcome these difficulties in robustly realizing the flight formation of QS systems become particularly important and deserve further exploration.

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The formation control of quadrotor is susceptible to the influence of complex external environments, thereby compromising the stability and safety of formation flight. The solutions for mitigating the impact of external disturbances on QS can be broadly categorized into two distinct approaches: designing disturbance observers and robust controllers. The fundamental concept of a disturbance observer is to estimate and compensate for system disturbances. In [12] and [13], a disturbance observer was designed to observe and compensate for unknown disturbances, improving formation control's stability and accuracy. However, the formation topology of these methods is relatively complex, which leads to a large amount of calculation. The main idea of a robust controller is to design controllers that can effectively suppress the effects of external disturbances. For instance, a robust controller was designed by integrating adaptive control, sliding mode control, and neural network techniques to effectively suppress external disturbances and achieve superior formation tracking performance [8], [14], [15]. Although robust controllers can resist external disturbances and uncertainties, they also have problems such as high complexity and difficulties in controller design and tuning. Among these robust control methods, adaptive control techniques are the most widely used. The  $\mathcal{L}_1$  adaptive control theory improves on the model reference adaptive control structure by deriving sufficient conditions for stability and performance based on the  $\mathcal{L}_1$ -paradigm of the underlying transfer function. In addition, the introduction of a low-pass filter to suppress the effect of the high-frequency response decouples the control law from the adaptive law, and the speed of adaptation can be arbitrarily increased, limited only by the hardware. Therefore,  $\mathcal{L}_1$  adaptive control can effectively mitigate the adverse effects of rapidly time-varying uncertainty on system response while ensuring robustness [16], [17], [18]. Based on this, the  $\mathcal{L}_1$  control technique is very suitable for the high-precision control of complex objects such as quadrotors. In [19], a geometric  $\mathcal{L}_1$  adaptive control law was proposed to estimate external disturbances, ensuring exponential stability of the nominal dynamics. However, only external disturbances of the attitude loop are considered. On this basis, the geometric  $\mathcal{L}_1$  adaptive control method was extended to the position loop [20]. In [21], a fast trajectory planning and robust trajectory tracking method for model predictive integral control combined with  $\mathcal{L}_1$  adaptive control was proposed. All of the above literature demonstrates the effectiveness of the  $\mathcal{L}_1$  adaptive approach to improve the safety and stability of quadcopters. However, these approaches only address matched disturbances, while the time-varying wind disturbances most commonly encountered by quadrotors are unmatched disturbances. Furthermore, there is also almost no literature that investigates the application of  $\mathcal{L}_1$  adaptive control in quadrotor formation control. Therefore, the problem of disturbance resistance in QS remains a valuable and open research topic.

The control of quadrotor formation faces challenges in obtaining the system's internal state, such as unmodelled dynamic parameters, unreliable velocity estimation, and variations in quadrotor mass. To address these issues, some studies combine adaptive, sliding mode, and other methods to design robust controllers that can handle the uncertainty of quadrotor

model parameters [14], [22]. It is widely acknowledged that the linear velocity of a quadrotor in quadrotor formation control cannot be directly measured by sensor, and even when data from multiple sensors are fused, significant errors may still occur. Consequently, several methods have been proposed to address this issue without velocity measurements. In [23], two auxiliary power systems driven by appropriate inputs were designed to overcome the deficiencies in linear and angular velocities respectively to achieve stable formation flight control. An enhanced fixed-time extended state observer was designed to estimate the speed of each quadrotor in the formation [24]. The literature [25] presented a distributed formation maneuvering control method for multi-agent systems, which was based on the complex Laplacian approach. In this method, the follower UAV achieves the control objective by relying solely on relative position information without requiring velocity information. However, the computation and formation topology of these methods are relatively complex. An arbitrary order differentiator was proposed in the literature [26], which is simple in structure, has the property of consistent finite-time accurate convergence. The differentiator can accurately estimate arbitrary order derivatives, and is particularly suitable for the estimation of quadrotor velocity information. On the other hand, changes in the quadrotor's overall mass due to energy consumption, load variations, etc. during the mission must be considered in controller design to ensure stable and safe flight. In [27], a multi-model adaptive estimator was designed to estimate the unknown mass and inertial momentum. Still, the estimation of the mass and other states was based on measurements from onboard sensors, which has some limitations. In [28], a backstepping-based fractional order sliding mode controller was designed to enable a quadrotor to track the desired path by introducing an adaptive correction factor for load estimation in its mass parameter. It should be noted that although some papers have studied the control problem of a single quadrotor under various loads, there is scarce literature addressing the inconsistent variation in mass among quadrotors during formation control. To our knowledge, no existing research simultaneously investigates the formation control problem for quadrotors without velocity feedback and with varying masses. Hence, it is crucial to study how to achieve effective formation control for quadrotors when accurate measurements of internal states like mass and velocity are challenging.

In this paper, a robust formation control framework is proposed that can overcome the dual challenges of disturbance from the external complex environment and unreliable state estimation within the QS system. Compared with previous research on quadrotor formation control, the main contributions of this paper are as follows

- 1) A distributed quadrotor formation control framework is proposed under a directed graph topology, based on a geometric controller, integrated with an  $\mathcal{L}_1$  adaptive controller and a differentiator, the robust formation control scheme without velocity feedback, can against disturbances and inconsistent mass changes.

- 2) The  $\mathcal{L}_1$  adaptive controllers are designed separately for the position and attitude loops and can deal with both matching

and unmatching disturbances. Unlike literature [19], [20], [21] which can only deal with matching disturbances. In addition, unlike the literature [23], [24], [25] assumes that all UAVs have equal mass, the inconsistent variation of the mass of each quadrotor in the QS system is considered.

3) The differentiator system is designed, and the two differentiators are utilized to estimate the linear velocity as well as the first and second derivatives of the virtual control force exerted on the quadrotor UAV. While [27], and [28] assume that the linear velocity can be obtained directly, and do not consider the inaccuracy and unreliability in the process of obtaining the linear velocity.

The sections in this article are organized as follows: Section II presents the system modeling and delineates the problem statement. In Section III, the robust formation controller based on the  $\mathcal{L}_1$  adaptive controller and differentiator design are given. The effectiveness of the proposed control method is demonstrated through simulation and experimental results presented in Section IV. Section V concludes this article.

## II. PROBLEM FORMULATION

### A. Preliminaries on Graph Theory

Consider a QS system comprising  $n$  interacting quadrotors.  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  is a directed communication topology network consisting of a finite number of vertices  $\mathcal{V} = \{v_1, v_2, v_3, \dots\}$  and the set of edges  $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ . In the graph  $\mathcal{G}$ , each vertex represents a quadrotor, and each edge  $(\mathcal{V}_i, \mathcal{V}_j)$  signifies that the  $i_{th}$  quadrotor can obtain information from its neighboring  $j_{th}$  quadrotor through sensing or communication. Let  $\mathcal{N}_i$  be the set of quadrotor dynamic neighborhoods that can interact with the  $i_{th}$  quadrotor. The adjacency matrix is expressed as  $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$ ,  $a_{ij}$  is the value of the communication topology between the  $i_{th}$  quadrotor and the  $j_{th}$  quadrotor, which equals 1 if the quadrotor  $i_{th}$  can communicate with the  $j_{th}$  quadrotor and 0 otherwise. In addition, the Laplacian matrix  $\mathcal{L} = [l_{ij}] \in \mathbb{R}^{n \times n}$  associated with  $\mathcal{G}$  is defined as follows

$$\mathcal{L} = \begin{cases} -a_{ij}, & \text{if } i \neq j \\ \sum_{j=1, j \neq i}^n a_{ij}, & \text{if } i = j \end{cases} \quad (1)$$

To ensure the formation's flexibility and scalability, we use the virtual leader-follower formation control strategy. Assuming that the virtual leader is marked as 0, the  $n$  followers are numbered and marked from 1 to  $n$ . Thus, the overall communication topology network of the QS is represented by the directed graph  $\tilde{\mathcal{G}} = (\tilde{\mathcal{V}}, \tilde{\mathcal{E}})$ , where  $\tilde{\mathcal{V}} = \{0, \mathcal{V}\}$ ,  $\tilde{\mathcal{E}} = \tilde{\mathcal{V}} \times \tilde{\mathcal{V}}$ . The adjacency matrix between followers and virtual leader is denoted as  $\mathcal{D} = \text{diag}(a_{10}, a_{20}, \dots, a_{n0})$ . If the virtual leader is a neighbor of follower  $i$ , we define  $a_{i0} = 1$ , otherwise  $a_{i0} = 0$ . Therefore, the directed graph  $\tilde{\mathcal{G}}$  communication topology under the virtual leader-follower formation strategy is represented by the interaction matrix  $\mathcal{M} = \mathcal{L} + \mathcal{D}$ .

### B. System Description and Modeling

Initially, establish the inertial coordinate system  $F_I$  and the body-fixed coordinate system  $F_B$ . The dynamics of each

quadrotor are modeled expressed as

$$\begin{cases} \dot{x}_i = v_i \\ \dot{v}_i = -\left(\frac{1}{m_{0i} + m_{\Delta i}}\right) f_i R_i e_3 + g e_3 + d_{fi} \\ \dot{R}_i = R_i \omega_i^\wedge \\ J_i \dot{\omega}_i = \tau_i - \omega_i \times J_i \omega_i + d_{\tau i} \end{cases} \quad (2)$$

where  $x_i \in \mathbb{R}^3$  and  $v_i \in \mathbb{R}^3$  denote the measurable location and the unmeasurable linear velocity, respectively.  $R_i \in \text{SO}(3)$  is the attitude,  $\omega_i \in \mathbb{R}^3$  is the angular velocity in the body-fixed frame,  $g$  is the constant acceleration of gravity,  $e_3 = [0; 0; 1]$  is constant vector,  $m_{0i}$  and  $m_{\Delta i}$  represents the quadrotors' standard mass and changing mass, respectively.  $J_i \in \mathbb{R}^{3 \times 3}$  is the inertia matrix,  $f_i \in \mathbb{R}$  and  $\tau_i \in \mathbb{R}^3$  represents total thrust and torque, respectively.  $d_{fi} \in \mathbb{R}^3$  is the lumped disturbance suffered in translation subsystem, including the unmodelled uncertainty parameters, external disturbances, among others.  $d_{\tau i} \in \mathbb{R}^3$  is the lumped disturbance in attitude subsystem that is composed of unknown parameters, external disturbances.  $(\cdot)^\wedge : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$  is defined as  $(a)^\wedge b = a \times b$  for all  $a, b \in \mathbb{R}^3$ , where  $\times$  denotes the cross product. The inverse of  $(\cdot)^\wedge : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$  is expressed as  $(\cdot)^\vee$ .

### C. Problem Statement

System (2) employs a virtual leader-follower control strategy and obviating the need for velocity information. Each follower quadrotor is equipped with inertial measurement units (IMU), ultra-wideband (UWB), radar altimeter to obtain localization information. The objective is to design a robust formation controllers for QS systems (2) to achieve

(1) distributed virtual leader-follower robust formation control without linear velocity measurements in directed graph communication topologies.

(2) the trajectory of the virtual leader is precisely tracked by each follower quadrotor, even in the presence of external disturbances and inconsistent mass changes.

The control objective can be described mathematically as, for  $\forall i, j = 1, \dots, n$ , satisfying:

$$\begin{cases} \lim_{t \rightarrow \infty} (x_i(t) - x_j(t)) = p_{ij} \\ \lim_{t \rightarrow \infty} (v_i(t) - \dot{x}_0(t)) = 0 \end{cases} \quad (3)$$

where  $p_{ij}$  indicates the relative position between quadrotor  $i$  and  $j$  which determines the desired formation pattern.  $x_0$  is the time-varying smooth trajectory of the virtual leader with second-order differentiable and bounded. According to the actual condition, make the following assumptions

*Assumption 1:* In the leader-follower graph  $\tilde{\mathcal{G}}$ , all of the followers are connected to the virtual leader, and a directed spanning tree can be found in the directed follower graph  $\mathcal{G}$ .

*Assumption 2:* The leader's trajectory  $x_0(t)$  is sufficiently smooth, second-order differentiable, and bounded.

*Assumption 3:* The external disturbance  $d_{fi}$  and  $d_{\tau i}$  are first-order differentiable and bounded, i.e.,  $\|d_{fi}(t)\| \leq \Delta_{df}$ ,  $\|d_{\tau i}(t)\| \leq \Delta_{\tau}$  for  $i \in \{1, \dots, n\}$ .

It is worth noting that  $d_{fi}$  and  $d_{\tau i}$  mainly include unmodelled parameters, and external wind disturbances, etc.,

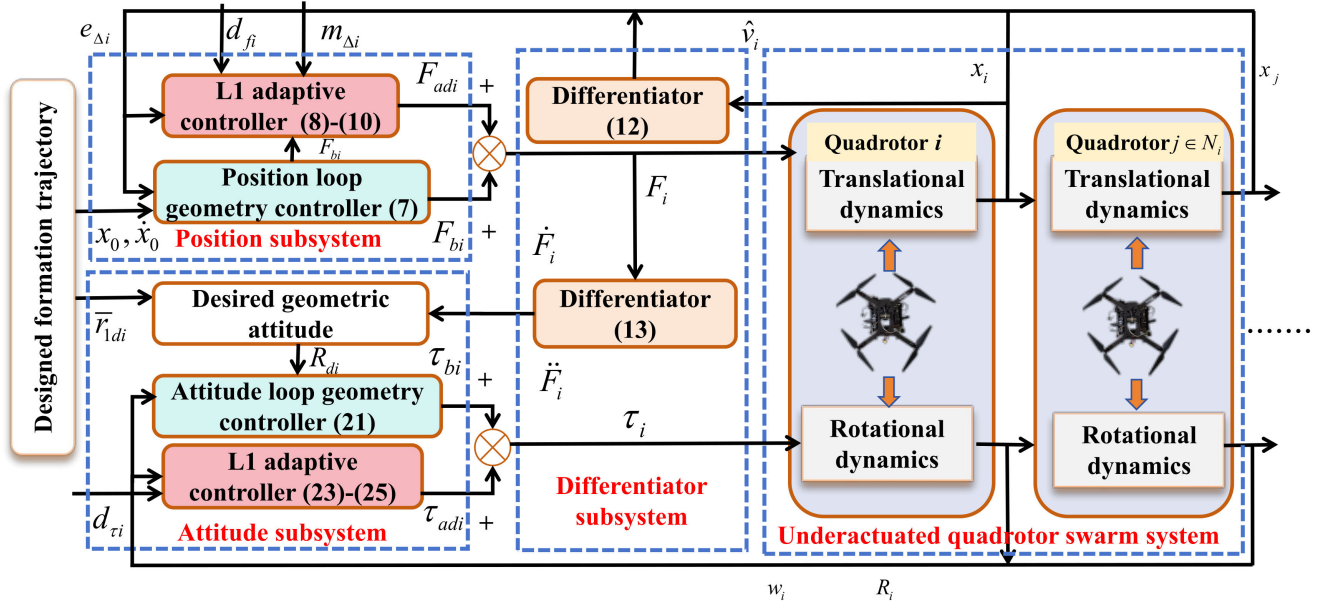


Fig. 1. Quadrotor UAV swarm formation control scheme.

generally bound in engineering. In practice, the motion of the quadrotor is subject to several engineering constraints that ensure the quadrotor motion trajectory is differentiable and bounded. Therefore, Assumptions 2 and 3 are reasonable.

### III. MAIN RESULT

Considering that the studied QS system is composed of underactuated quadrotors with strong coupling nonlinearity. To decouple the position loop and attitude loop, the QS is divided into an inner loop and an outer loop cascade structure, so as to simplify the control design and improve the stability and accuracy of the control. A geometric controller without linear velocity measurement is designed to achieve the tracking control of the formation, and an  $\mathcal{L}_1$  adaptive algorithm is designed to identify and estimate the internal and external uncertainties. Fig. 1 represents the general scheme diagram of formation control.

#### A. $\mathcal{L}_1$ Adaptive Based Formation Tracking Controller Design

In this subsection, a distributed robust formation position controller is designed to implement the control of each quadrotor position, velocity, and acceleration. Firstly, the tracking error of each quadrotor with the virtual leader is defined as follows

$$\begin{aligned} e_{xi} &= x_i - x_0 - p_i \\ e_{vi} &= v_i - \dot{x}_0 \end{aligned} \quad (4)$$

where  $p_i$  denotes the reference relative position of the  $i_{th}$  follower quadrotor with respect to the virtual leader,  $e_{xi}$  and  $e_{vi}$  denote the position and velocity tracking errors of the  $i_{th}$  follower quadrotor concerning the virtual leader, respectively.

The formation error  $\Delta_{ei} = [\Delta_{xi}^T \Delta_{vi}^T]^T$  can be described by the following mathematical form

$$\begin{aligned} \Delta_{xi} &= \sum_{j \in \mathcal{N}_i} a_{ij}(x_i - x_j - p_{ij}) + a_{i0}e_{xi} \\ \Delta_{vi} &= \sum_{j \in \mathcal{N}_i} a_{ij}(v_i - v_j) + a_{i0}e_{vi} \end{aligned} \quad (5)$$

where  $\Delta_{xi}$  and  $\Delta_{vi}$  denote the formation position error and velocity error, respectively.  $p_{ij}$  denotes the desired relative position between each follower's quadrotors.

To enhance the formation tracking performance of the translation subsystem, the intermediate translation control signal  $F_i$  is constructed as two components as

$$F_i = F_{bi} + F_{adi} \quad (6)$$

where  $F_{bi} = -f_i R_i e_3$  is the nominal virtual control input.  $F_{adi}$  is the  $\mathcal{L}_1$  adaptive controller employed for estimation unknown disturbances and uncertain states.

The nominal virtual control law is designed as

$$F_{bi} = m_{0i}(-k_x \Delta_{xi} - k_v \Delta_{vi} - g e_3 + \ddot{x}_0) \quad (7)$$

where  $k_x$  and  $k_v$  are positive paramaters.

The  $\mathcal{L}_1$  adaptive control law comprises a state predictor, an adaptive law, and a low-pass filter (LPF). This controller offers the advantages of robustness and anti-interference capability, along with favorable response speed and fast online parameter estimation. Consequently, it is well-suited for achieving high-precision control of complex controlled objects such as quadrotors.

The state predictor has a consistent mathematical expression with the plant object, which helps to verify whether the estimated uncertainty is correct. If the uncertainty estimated by the adaptation law is accurate, the prediction error should



be zero. It is expressed as

$$\dot{\hat{v}}_{li} = g e_3 + \frac{F_{bi}}{m_{0i}} + F_{bi} \hat{\Delta}_{mi} + \hat{d}_{fi} + F_{adi} + A_{ps} \tilde{v}_{li} \quad (8)$$

where  $\tilde{v}_{li} = \hat{v}_{li} - v_i$  denotes predictor error,  $A_{ps} \in \mathbb{R}^{3 \times 3}$  is the chosen diagonal Hurwitz matrix, which serves to make the prediction error  $\tilde{v}_{li}$  converge exponentially fast to 0.  $\hat{\Delta}_{mi}$  and  $\hat{d}_{fi}$  are the estimates of the uncertainties states  $\Delta_{mi} = 1/(m_{0i} + m_{\Delta i}) - 1/m_{0i}$  and  $d_{fi}$ , respectively.

The piecewise-constant adaptive estimation law is designed to estimate the mass variation of the quadrotor and the lumped disturbance of the position loop. At period  $t \in [bT_s, (b+1)T_s]$ , its mathematical form is expressed as

$$\begin{cases} F_{bi} \hat{\Delta}_{mi}(t) = F_{bi} \hat{\Delta}_{mi}(bT_s) = -\Phi^{-1} \mu(bT_s) \\ \hat{d}_{fi}(t) = \hat{d}_{fi}(bT_s) = -\Phi^{-1} \mu(bT_s) \end{cases} \quad (9)$$

where  $\Phi^{-1} = A_{ps}^{-1}(\exp(A_{ps}T_s) - I_3)$  and  $\mu(bT_s) = \exp(A_{ps})\tilde{v}_{li}(bT_s)$ ,  $b$  and  $T_s$  are the time order and time step, respectively.

The position loop  $\mathcal{L}_1$  adaptive control law is designed as

$$F_{adi} = -C_p(s)(F_{bi} \hat{\Delta}_{mi} + \hat{d}_{fi}) \quad (10)$$

where  $C_p(s) = w_p/(s+w_p)$  is the LPF with  $w_p$  being the filter cut-off frequency which can effectively mitigate the impact of high-frequency oscillations caused by the high adaptive gain such that the quadrotor controller has good adaptive performance and robustness simultaneously.

The net thrust input for each quadrotor is obtained as

$$f_i = -F_i \cdot R_i e_3 \quad (11)$$

where operators  $(\cdot)$  denotes the inner product of vectors.

### B. Differentiator Systems

The distributed robust formation controller proposed in this paper relies on the position and velocity information of the quadrotor UAV. In practical quadrotor applications, direct measurement of the velocity using sensors is not available, and fusing data from other sensors introduces a specific error in the obtained velocity. Additionally, in practical measurements, there will be small high-frequency input noise that affects the higher-order derivatives of the position signal. This interference introduces inaccuracies in the velocities obtained using the difference method. Therefore, we use the exact differentiator in the literature [26] to design a higher-order sliding mode (HOSM) differentiator to estimate the velocity of each quadrotor in a QS system. Eventually, we can get the position controller based on the differentiator that rely only on position information without velocity measurements. The differentiator is stated as

$$\begin{aligned} \dot{\hat{x}}_i &= -\rho_1 \vartheta |\tilde{x}_i|^{\frac{1}{2}} \text{sign}(\tilde{x}_i) - \rho_2 (1 - \vartheta) |\tilde{x}_i|^{\frac{2+\gamma}{2}} \text{sign}(\tilde{x}_i) + \hat{v}_i \\ \dot{\hat{v}}_i &= -\rho_1 \vartheta \text{sign}(\tilde{x}_i) - \rho_2 (1 - \vartheta) |\tilde{x}_i|^{1+\gamma} \text{sign}(\tilde{x}_i) \end{aligned} \quad (12)$$

where  $\hat{x}_i$  is the estimated value of  $x_i$  and  $\tilde{x}_i = x_i - \hat{x}_i$  is the differentiator error of the position estimate.  $\rho_1 > 0$ ,  $\rho_2 > 0$ ,  $\gamma > 0$  are positive constants gains. Select an arbitrary positive number  $T_b$ , when  $t < T_b$ ,  $\vartheta = 0$ , otherwise

$\vartheta = 1$ . As demonstrated in [26], the dynamics system (12) is uniformly finite-time exact convergent. Therefore, we can obtain an accurate estimate  $\hat{v}_i$  of the linear velocity of the quadrotor through the differentiator. Further, we can obtain an accurate estimate of  $\hat{\Delta}_{vi}$ . Eventually, we can get the position controller based on the differentiator without velocity measurement. The utilization of a differentiator for linear velocity estimation diminishes the reliance on supplementary sensors, streamlines the system architecture, reduces hardware and software intricacy, minimizes system costs, and enhances both simplicity and reliability.

*Remark 1:* Since the designed differentiator is consistently finite time exactly convergent, the separation principle is satisfied. Therefore, it's designed separately from the controller to simplify the stability proof process of the control system.

Considering that the desired angular velocity and its first-order derivative required for designing the geometric attitude controller is dependent on the first-order and second-order derivatives of the desired attitude, which in turn rely on  $\dot{F}_i$  and  $\ddot{F}_i$ . Since determining  $\dot{F}_i$  and  $\ddot{F}_i$  using the difference method is computationally intensive and prone to significant errors. Therefore, the design of an arbitrary order differentiator for estimating the first and second derivatives of virtual thrust not only reduces computational complexity and simplifies controller design but also enhances control accuracy and flight performance.

Define the estimate of  $F_i$  as  $\hat{F}_i$ , then the estimation error of  $F_i$  concerning the differentiator is  $\tilde{F}_i = F_i - \hat{F}_i$ . The form of the differentiator is expressed as follows

$$\begin{aligned} \dot{\hat{F}}_i &= -\rho_3 \vartheta |\tilde{F}_i|^{\frac{2}{3}} \text{sign}(\tilde{F}_i) - \rho_4 (1 - \vartheta) |\tilde{F}_i|^{\frac{3+\gamma}{3}} \text{sign}(\tilde{F}_i) + \hat{F}_i \\ \dot{\hat{F}}_i &= -\rho_3 \vartheta |\tilde{F}_i|^{\frac{1}{3}} \text{sign}(\tilde{F}_i) - \rho_4 (1 - \vartheta) |\tilde{F}_i|^{\frac{3+2\gamma}{3}} \text{sign}(\tilde{F}_i) + \hat{F}_i \\ \dot{\hat{F}}_i &= -\rho_3 \vartheta \text{sign}(\tilde{F}_i) - \rho_4 (1 - \vartheta) |\tilde{F}_i|^{1+\gamma} \text{sign}(\tilde{F}_i) \end{aligned} \quad (13)$$

The parameter selection in (13) is identical to that in (12). Similarly, the system (13) is consistently finite time exactly convergent. Therefore, the estimation  $F_i$  of both first-order and second-order derivatives can be conducted. This is needed to design the attitude controller in our later subsection.

### C. $\mathcal{L}_1$ Adaptive Based Geometric Attitude Controller Design

In this subsection, the robust attitude controllers are designed for QS that are resistant to internal and external uncertainties. In the first step, the geometric attitude controllers is designed on the nonlinear manifold  $\text{SO}(3)$ . In the second step, the  $\mathcal{L}_1$  adaptive control law is designed to recognize and compensate for the uncertainties in the attitude loop such that the actual attitude  $R_i$  tracks the desired attitude  $R_{di}$ .

Firstly, the desired attitude  $R_{di}$  is calculated for the period based on the previously designed position loop intermediate control variable  $F_i$ . Since  $r_{3di}$  is a column of the rotation matrix that satisfies  $r_{3di}^T r_{3di} = 1$ , therefore,

$$r_{3di} = \frac{F_i}{\|F_i\|} \quad (14)$$

Then, a yaw angle  $\psi_d$  is chosen, corresponding to the expectation vector  $\bar{r}_{1di} = [\cos \psi_d, \sin \psi_d, 0]^T$ , which is not

parallel to  $r_{3di}$ . The corresponding  $r_{2di}, r_{3di}$  are calculated by the following expression,

$$r_{2di} = \frac{r_{3di} \times \bar{r}_{1di}}{\|r_{3di} \times \bar{r}_{1di}\|}, r_{1di} = r_{2di} \times r_{3di} \quad (15)$$

Ultimately, one can obtain  $R_{di}$  as

$$R_{di} = [r_{1di}, r_{2di}, r_{3di}]^T \quad (16)$$

After decoupling the desired attitude, the attitude controller needs to be designed to make each follower quadrotor track the desired attitude  $R_{di}$ . It ensures that the attitude tracking error of the control system converges quickly to a bound, so that the position tracking error also converges quickly to a bound, so as to realize the stability of the whole dynamics.

The corresponding desired angular velocity is expressed as

$$\omega_{di} = (R_{di}^T \dot{R}_{di})^\vee \quad (17)$$

The rotation matrix error is denoted as  $R_{ei} = R_i^T R_{di}$ . As defined in Section IV of the literature [29], the error function on the Lie group SO(3) is defined as

$$\Psi_i(R_i, R_{di}) = \frac{1}{2} \text{tr}[I_3 - R_{di}^T R_i] \quad (18)$$

where  $\text{tr}(\cdot)$  denotes the trace of a square matrix.

The corresponding attitude-tracking error is defined as

$$e_{Ri} = \frac{1}{2} (R_{di}^T R_i - R_i^T R_{di})^\vee \quad (19)$$

Furthermore, the angular velocity error is defined as

$$e_{\omega i} = \omega_i - R_{ei} \omega_{di} \quad (20)$$

Eventually, combining the first-order and second-order derivatives of  $F_i$  estimated by the differentiator system (13) to obtain  $\bar{w}_{di}$  and  $\dot{\bar{w}}_{di}$ , we design the rotation matrix-based geometric attitude controller following as

$$\begin{aligned} \tau_{bi} = & -k_R e_{Ri} - k_w e_{\omega i} + \omega_i \times J_i \omega_i \\ & - J_i (\omega_i^{\wedge} R_i^T R_{di} \bar{w}_{di} - R_i^T R_{di} \dot{\bar{w}}_{di}) \end{aligned} \quad (21)$$

where  $k_R$  and  $k_w$  are the positive parameters.

The geometric controller ensures exponential stability of the system. However, when there are internal and external uncertainties in the system such as unknown parameters, external disturbances, the stability of the system deteriorates. Therefore, designing an attitude loop  $\mathcal{L}_1$  adaptive control law recognizes and compensates for internal and external uncertainties.

The dynamic equation for angular velocity in the QS system (2) is rewritten as

$$\dot{\omega}_i = -J_i^{-1}(\omega_i \times J_i \omega_i) + J_i^{-1}(\tau_{bi} + \tau_{adi} + d_{\tau i}) \quad (22)$$

where  $\tau_{adi}$  is the  $\mathcal{L}_1$  adaptive control law to be designed later.

The state predictor of the attitude loop is

$$\dot{\hat{\omega}}_i = -J_i^{-1}(\omega_i \times J_i \omega_i) + J_i^{-1}(\tau_{bi} + \tau_{adi} + \hat{d}_{\tau i}) + A_{as} \tilde{\omega}_i \quad (23)$$

where  $\tilde{\omega}_i = \hat{\omega}_i - \omega_i$  is the prediction error.  $A_{as} \in \mathbb{R}^{3 \times 3}$  is our chosen diagonal Hurwitz matrix, which serves to make the prediction error  $\tilde{\omega}_i$  converge exponentially fast to zero.

Then, at time period  $t \in [bT_s, (b+1)T_s]$ , the design of the segmentation constant adaptive estimation law is

$$\hat{d}_{\tau i}(t) = \hat{d}_{\tau i}(bT_s) = -J_i \Phi^{-1} \mu(bT_s) \quad (24)$$

where  $\Phi^{-1} = A_{as}^{-1}(\exp(A_{as} T_s) - I_3)$  and  $\mu(aT_s) = \exp(A_{as}) \tilde{\omega}_i(bT_s)$ .

The attitude loop  $\mathcal{L}_1$  adaptive control law is designed as

$$\tau_{adi} = -C_a(s) \hat{d}_{\tau i} \quad (25)$$

where  $C_a(s) = w_a/(s + w_a)$  is the LPF to prevent high-frequency components from entering the control channel, and  $w_a$  is the filter cut-off frequency. Therefore, the final control torque for each quadrotor in the QS system is

$$\tau_i = \tau_{bi} + \tau_{adi} \quad (26)$$

#### D. Stability Analysis

*Theorem 1:* Consider the dynamics of the QS (2) under Assumptions 1, 2, and 3, employing  $\mathcal{L}_1$  adaptive control-based geometry control law, where thrust is provided by (11) and input torque is provided by (26). Then the formation error and attitude error of each quadrotor is uniformly bounded and quickly achieve the desired formation pattern.

*Proof:* The stability of the QS is demonstrated through a two-step demonstration.

*Step 1:* Proving the stability of the attitude subsystem. The time derivatives of  $e_{Ri}$  and  $e_{\omega i}$  can be written respectively as

$$\begin{aligned} \dot{e}_{Ri} &= \frac{1}{2} (R_{di}^T R_i \hat{e}_{\omega i} + \hat{e}_{\omega i} R_i^T R_{di})^\vee \\ &= C_i(R_{di}^T R_i) e_{\omega i}. \end{aligned} \quad (27)$$

$$\dot{e}_{\omega i} = J_i^{-1}(\tau_i + d_{\tau i}) - H_i \quad (28)$$

where  $C_i(R_{di}^T R_i)$  and  $H_i$  are respectively given by

$$\begin{aligned} C_i(R_{di}^T R_i) &= \frac{1}{2} (\text{tr}[R_{di}^T R_i] I_3 - R_i^T R_{di}) \\ H_i &= J_i^{-1}(\omega_i \times J_i \omega_i) - \omega_i^{\wedge} R_{ei} \omega_{di} + R_{ei} \dot{\omega}_{di} \end{aligned}$$

Substituting (26) into (28) gives

$$J_i \dot{e}_{\omega i} = -k_R e_{Ri} - k_w e_{\omega i} + E_i \quad (29)$$

where  $E_i = \tilde{F}_i + e_{d\tau i}$  with  $e_{d\tau i} = d_{\tau i} - \tau_{adi}$  being the disturbance estimation error of the  $\mathcal{L}_1$  adaptive control estimation law. The Lyapunov candidate function for the rotational subsystem is chosen to be

$$V_a = \frac{1}{2} e_{\omega i} \cdot J_i e_{\omega i} + k_R \Psi(R_i, R_{di}) + c_2 e_{Ri} \cdot e_{\omega i} \quad (30)$$

where  $c_2$  is a positive constant.

According to [31, Appendix B], the following relationships hold

$$\|\dot{e}_{Ri}\| \leq \|e_{\omega i}\|. \quad (31)$$

$$\frac{1}{2} \|e_{Ri}\|^2 \leq \Psi_i(R_i, R_{di}) \leq \frac{1}{2 - \psi_i} \|e_{Ri}\|^2 \quad (32)$$

$$\dot{\Psi}_i(R_i, R_{di}) = \frac{1}{2} e_{\omega i}^T (R_{di}^T R_i - R_i^T R_{di})^\vee = e_{Ri} \cdot e_{\omega i} \quad (33)$$

where  $\psi_i \in \mathbb{R}^+$  satisfies  $\Psi_i(R_i, R_{di}) \leq \psi_i < 2$ . According (30) and  $\lambda_{\min}(J_i) \|e_{\omega i}\|^2 \leq e_{\omega i} \cdot J_i e_{\omega i} \leq \lambda_{\max}(J_i) \|e_{\omega i}\|^2$ ,

where  $\lambda_{\min}$  and  $\lambda_{\max}$  are the minimum and maximum eigenvalues, respectively, of  $J_i$ . This shows that  $\Psi_i$  is positive definite and decrescent. Consequently, the  $V_a$  is bounded as

$$z_a^T M_{a1} z_a \leq V_a \leq z_a^T M_{a2} z_a \quad (34)$$

where  $z_a = [\|e_{Ri}\|, \|e_{\omega i}\|]^T \in \mathbb{R}^2$  and

$$M_{a1} = \frac{1}{2} \begin{bmatrix} k_R & -c_2 \\ -c_2 & \lambda_{\min}(J_i) \end{bmatrix}, \quad M_{a2} = \frac{1}{2} \begin{bmatrix} \frac{2k_R}{2-\psi_i} & c_2 \\ c_2 & \lambda_{\max}(J_i) \end{bmatrix}$$

$c_2$  satisfied as

$$c_2 < \min \left\{ k_\omega, \frac{4k_\omega k_R \lambda_{\min}(J_i)^2}{k_\omega \lambda_{\max}(J_i) + 4k_R \lambda_{\min}(J_i)^2}, \sqrt{k_R \lambda_{\min}(J_i)} \right\}$$

The time derivative of  $V_a$  is given by

$$\begin{aligned} \dot{V}_a &= e_\omega \cdot J_i \dot{e}_{\omega i} + k_R e_{Ri} \cdot e_{\omega i} + c_2 \dot{e}_{Ri} \cdot e_\omega + c_2 e_{Ri} \cdot \dot{e}_{\omega i} \\ &= -k_\omega \|e_{\omega i}\|^2 - c_2 k_R e_{Ri} \cdot J_i^{-1} e_{Ri} + c_2 C_i(R_{di}^T R_i) e_{\omega i} \cdot e_{\omega i} \\ &\quad - c_2 k_\omega e_{Ri} \cdot J_i^{-1} e_{\omega i} + e_{\omega i} \cdot E_{d\tau i} \end{aligned} \quad (35)$$

Since  $\|\dot{e}_{Ri}\| \leq \|e_{\omega i}\|$  and  $\|C_i(R_{di}^T R_i)\| \leq 1$ , by applying Young's inequality, we can derive

$$e_{\omega i} \cdot E_{d\tau i} \leq \ell \|e_{\omega i}\|^2 + \frac{1}{4\ell} \|E_{d\tau i}\|^2 \quad (36)$$

Use (29),(31) and (33),(36), the  $\dot{V}_a$  can be rewritten as

$$\dot{V}_a \leq -z_a^T W_a z_a + \frac{1}{4\ell} \|E_{d\tau i}\|^2 \quad (37)$$

where  $\ell$  is a positive constant and the matrix  $W_a \in \mathbb{R}^{2 \times 2}$  is given by

$$W_a = \begin{bmatrix} \frac{c_2 k_R}{\lambda_{\max}(J_i)} & -\frac{c_2 k_\omega}{2\lambda_{\min}(J_i)} \\ -\frac{c_2 k_\omega}{2\lambda_{\min}(J_i)} & k_\omega - c_2 - \ell \end{bmatrix} \quad (38)$$

Now we show that  $E_{d\tau i} = \tilde{F}_i + e_{d\tau i}$  is bounded. Angulo and Moreno [26] proved that the arbitrary order differentiator error converges to zero in finite time, i.e.,  $\tilde{F}_i$  converges to zero in finite time.  $e_{d\tau i} = d_{\tau i} - \tau_{adi}$  is the disturbance estimation error of the  $\mathcal{L}_1$  adaptive control estimation law. To prove that  $\|\tau_{adi}\|$  is bounded, we need to prove  $\|\hat{d}_{\tau i}(bT_s)\|$  is bounded for all  $b \in \mathbb{N}$ . We employ inductive proof. When  $t = 0$ ,  $\|\hat{d}_{\tau i}(0)\| = 0$ . Since  $A_{as}$  is a Hurwitz matrix, the prediction error dynamics  $\tilde{w}_i(t) = A_{as} \tilde{w}_i(t) + J_i^{-1}(t)(d_{\tau i}(t) - \hat{d}_{\tau i}(t))$  is bounded-input, bounded-output stable. Using the bounds on  $d_{\tau i}(t)$  for  $t \in [bT_s, (b+1)T_s)$  in Assumption 3, it's concluded that  $\tilde{w}_i((b+1)T_s)$  is bounded. Considering the adaptation law in (24), it's concluded that  $\hat{d}_{\tau i}((b+1)T_s)$  is bounded. Thus,  $\|\hat{d}_{\tau i}(bT_s)\|$  is bounded for all  $b \in \mathbb{N}$ . Define  $\Delta_{\hat{d}_\tau}$  such that

$$\|\hat{d}_{\tau i}(bT_s)\| \leq \Delta_{\hat{d}_\tau} \quad (39)$$

for all  $b \in \mathbb{N}$ . Since  $C_a(s) = \omega_a/(s + \omega_a)$  with DC gain equal to 1, we have  $\|C_a(s)\|_{\mathcal{L}_1} = 1$ . Following [20, Lemma A.7.1], we obtain  $\|\tau_{adi}(t)\| \leq \|C_a(s)\|_{\mathcal{L}_1} \|\hat{d}_{\tau i}(t)\|_{\mathcal{L}_\infty} \leq \Delta_{\hat{d}_\tau}$ . Combine this with the fact  $d_{\tau i}$  is bounded in Assumption 3, so  $e_{d\tau i}$  is bounded. Furthermore, it's concluded that  $E_{d\tau i}$  is bounded by  $\bar{\Delta}_{d\tau} = \Delta_{d\tau} - \Delta_{\hat{d}_\tau}$ .  $\dot{V}_a$  is bounded by

$$\dot{V}_a \leq -z_a^T W_a z_a + \frac{1}{4\ell} \bar{\Delta}_{d\tau}^2 \quad (40)$$

Finally, using the Lyapunov stability theory, the attitude error  $e_{Ri}$  and the angular velocity error  $e_{\omega i}$  converge to the neighborhood of zero when  $t \rightarrow \infty$ .

*Remark 2:* The estimation error converges to the neighborhood of zero as the sampling time  $T_s$  goes to zero. The uncertainty estimation can be made more accurate by reducing  $T_s$ .

*Step 2:* Proving the stability of the tracking error dynamics. Let  $e_x = [e_{x1}^T, \dots, e_{xn}^T]^T$ ,  $e_v = [e_{v1}^T, \dots, e_{vn}^T]^T$ ,  $\Delta_m = [\Delta_{m1}^T, \dots, \Delta_{mn}^T]^T$ ,  $F_b = [F_{b1}^T, \dots, F_{bn}^T]^T$ ,  $d_f = [d_{f1}^T, \dots, d_{fn}^T]^T$ ,  $\Delta_v = [\Delta_{v1}^T, \dots, \Delta_{vn}^T]^T$ ,  $\mathcal{T} = [\mathcal{T}_1^T, \dots, \mathcal{T}_n^T]^T$  is a coupling term between the position and attitude loops. Define the position loop error vector consisting of position error and velocity error as

$$e_\Delta = [e_x^T \ e_v^T]^T \quad (41)$$

Combining (4), (41) and taking the time derivative of  $e_\Delta$ , one gets

$$\dot{e}_\Delta = (A_p \otimes I_3) e_\Delta + (B_p \otimes I_3) (\tilde{\Delta}_m F_b + \tilde{d}_f + \tilde{\Delta}_v + \mathcal{T}) \quad (42)$$

where  $A_p = \begin{bmatrix} 0_{n \times n} & I_{n \times n} \\ -k_x \mathcal{M} & -k_v \mathcal{M} \end{bmatrix} \in \mathbb{R}^{2n \times 2n}$ ,  $B_p = \begin{bmatrix} 0_{n \times n} & I_{n \times n} \end{bmatrix}^T \in \mathbb{R}^{2n \times n}$ ,  $\tilde{\Delta}_m = \Delta_m - \hat{\Delta}_m$  and  $\tilde{d}_f = d_f - \hat{d}_f$  are the  $\mathcal{L}_1$  adaptive law estimation errors of  $\Delta_{mi}$  and  $d_{fi}$ , respectively. Since  $A_{ps}$  is a Hurwitz matrix, the selected LPF satisfies  $C_p(s) = \omega_p/(s + \omega_p)$  with DC gain equal to 1 and  $\|C_p(s)\|_{\mathcal{L}_1} = 1$ . The prediction error dynamics  $\tilde{v}_{li} = A_{ps} \tilde{v}_{li} + I_3(F_{bi}(\Delta_{mi} - \hat{\Delta}_{mi}) + d_{fi} - \hat{d}_{fi})$  is bounded-input, bounded-output stable. There exists the reasonable assumption that  $d_{fi}$  and  $\Delta_{mi}$  are bounded. it's concluded that  $\tilde{v}_{li}$  are bounded. Based on the adaptive law in (9), it's concluded that  $\hat{\Delta}_{mi}$  and  $\hat{d}_{fi}$  are bounded. Therefore, the  $\hat{\Delta}_m$  and  $\hat{d}_f$  are bounded. Further, it can be concluded that  $\tilde{\Delta}_m$  and  $\tilde{d}_f$  converge to the neighborhood of zero when  $t \rightarrow \infty$ . On the other hand,  $\tilde{\Delta}_v = \Delta_v - \hat{\Delta}_v$  is the estimation error due to the system of differentiators (12), which has been shown in [26] differentiators errors to converge to 0 in finite time.  $\mathcal{T}$  have been proven to be bounded when  $t \rightarrow \infty$  in [29]. By employing the Lyapunov stability theory in cascade systems [30], it can be demonstrated that the closed-loop system exhibits bounded stability when the nominal system  $\dot{e}_\Delta = A_p e_\Delta$  is asymptotically stable. The eigenvalues of  $A_p$  are given by  $\lambda_{i\pm} = \frac{k_v \mu_i \pm \sqrt{k_v^2 \mu_i^2 + 4k_x \mu_i}}{2}$  where  $\mu_i$  denotes the  $i_{th}$  eigenvalue of  $-\mathcal{M}$ ,  $\lambda_i$  is the eigenvalues of  $A_p$ .

It can be seen that  $A_p$  has  $2k$  zero eigenvalues if and only if  $-\mathcal{M}$  has  $k$  zero eigenvalues. Base on Lemma 2.4 in [32],  $-\mathcal{M}$  has at least one zero eigenvalue with associated eigenvector  $\mathbf{1}_n$ , and all of the nonzero eigenvalues of  $-\mathcal{M}$  have negative real parts. Therefore, the  $A_p$  has at least two zero eigenvalues. Combining Assumption 1 with Lemma 4.1 in [32], The conclusion is that the nominal system  $\dot{e}_\Delta = A_p e_\Delta$  exhibits asymptotically stable.

Consequently,  $e_\Delta$  converges to the neighborhood of zero, which can guarantee that each quadrotor reaches the desired formation configuration. This completes the proof.

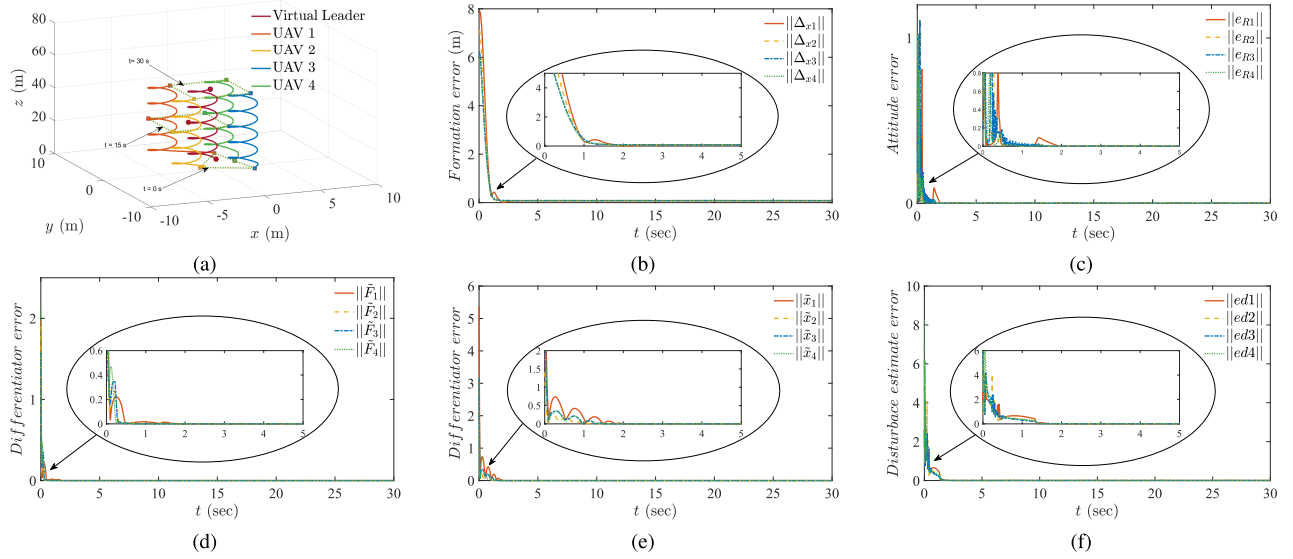


Fig. 2. Simulation results under mass variation and time-varying disturbances: (a) Trajectory of a QS system in 3D space. (b) Formation errors of each follower quadrotor. (c) Attitude error of each follower quadrotor. (d) Differentiator estimation error for virtual control inputs. (e) Differentiator estimation error of quadrotor position for each follower. (f) The estimation error based on the  $\mathcal{L}_1$  adaptive law is computed by

$$ed_i = \sqrt{\sum_{q=1}^N ((F_{bi} \Delta_{mi,q} - \hat{F}_{bi} \Delta_{mi,q})^2 + (d_{fi,q} - \hat{d}_{fi,q})^2 + (d_{ti,q} - \hat{d}_{ti,q})^2 / N)}, \text{ where } N \text{ denotes the total number of samples in the simulation.}$$

TABLE I  
FORMATION CONTROLLER AND DIFFERENTIATOR PARAMETERS

| param. | value | param.   | value           |
|--------|-------|----------|-----------------|
| $k_x$  | 3     | $A_{ps}$ | -diag(5,5,5)    |
| $k_v$  | 1.1   | $A_{as}$ | -diag(10,10,10) |
| $k_R$  | 5     | $\rho_1$ | 10              |
| $k_w$  | 0.15  | $\rho_2$ | 10              |
| $w_p$  | 10    | $\rho_3$ | 10              |
| $w_a$  | 6     | $\rho_4$ | 1               |
| $T_s$  | 0.002 | $\gamma$ | 0.006           |

#### IV. SIMULATION AND EXPERIMENTAL RESULTS

In this section, the validity of the proposed robust control approach under unreliable internal state acquisition and complex external time-varying disturbances are verified by numerical simulations and experiments. For descriptive purposes, all data that follows defaults to international standard units. The physical parameters of each follower quadrotor set as follows:  $m_i = 2$  for  $i = 1, 2, 3, 4$ , rotational inertia  $J_i = \text{diag}([3.12 \times 10^{-2}, 3.44 \times 10^{-2}, 6.21 \times 10^{-2}])$ . The initial position of each quadrotor is set as  $x_0 = [0, 0, 6]^T$ ,  $x_1 = [2, 5, 0]^T$ ,  $x_2 = [-1, 1, 0]^T$ ,  $x_3 = [3, -1, 0]^T$ ,  $x_4 = [2.5, 2, 0]^T$ . The relative positions between the 4 follower quadrotors and the virtual leader determine the desired formation of the formation, set as  $p_1 = [-2.5, 2.5, 0]^T$ ,  $p_2 = [-2.5, -2.5, 0]^T$ ,  $p_3 = [2.5, -2.5, 0]^T$ ,  $p_4 = [2.5, 2.5, 0]^T$ . The parameters of the base geometry controller,  $\mathcal{L}_1$  adaptive controller and differentiator are listed in table I.

##### A. Simulation Results for Mass Variation and Disturbances

Initial values of linear and angular velocities are set to 0. A virtual leader is the reference linear velocity set to  $\dot{x}_0 = [-0.8 \sin(t), 0.8 \cos(t), 1.2]^T$ , desired yaw angle is  $\psi_d = 0$ . The mass of each UAV is varied as  $m_{\Delta 1} = 0.31$ ,  $m_{\Delta 2} = 0.42$ ,  $m_{\Delta 3} = 0.15$ ,  $m_{\Delta 4} = 0.1$ . It is assumed that the

QS system is subjected to lumped disturbances including unmodelled dynamics, external wind disturbances, and aerodynamic effects between the quadrotors denoted as

$$d_{fi} = [0.5 \sin(0.2 t) + 0.25 \sin(0.5 t) + 1 \sin(t), 0.3 \sin(0.3 t) + 0.2 \sin(0.6 t) + 0.4 \sin(0.8 t), 0.8 \sin(2t) + 1 \sin(0.25 t)]^T,$$

$$d_{ti} = [0.7 \sin(0.3 t) + 0.5 \sin(0.75 t), 0.4 \sin(0.8 t) + 0.4 \sin(0.5 t) + 0.2 \sin(0.6 t), 0.8 \sin(2t) + 1 \sin(0.25 t)]^T$$

$d_{fi}$  and  $d_{ti}$  are a linear superposition of real-world disturbances of various frequencies, where  $d_{fi}$  contains unmatched disturbances. The results of the simulation is shown in Fig. 2. Fig. 2(a) plots a 3D trajectory map of the four followers, providing snapshots of the quadrotor positions every 15 s. It can be seen that the shape of the formation quickly reaches the desired square shape from the original triangle. The formation and tracking errors for each follower quadrotor are depicted in Fig. 2(b) and Fig. 2(c), respectively. It can be seen that it can quickly converge to near zero, proving that the proposed formation controller has better transient and steady-state performance. Fig. 2(d) and Fig. 2(e) shows the estimation errors of the differentiator systems (12), and (13), respectively. Fig. 2(f) depicts that the error between the estimated lumped disturbance (mass variation and time-varying external disturbance) and the actual disturbance of the  $\mathcal{L}_1$  controller quickly reaches 0. It shows that the designed  $\mathcal{L}_1$  controller accurately identifies the various uncertainties and compensates them quickly. Under various internal uncertainties and external disturbances, the simulation results demonstrate the exceptional performance of the proposed formation control scheme including a geometric controller and an  $\mathcal{L}_1$  adaptive controller.



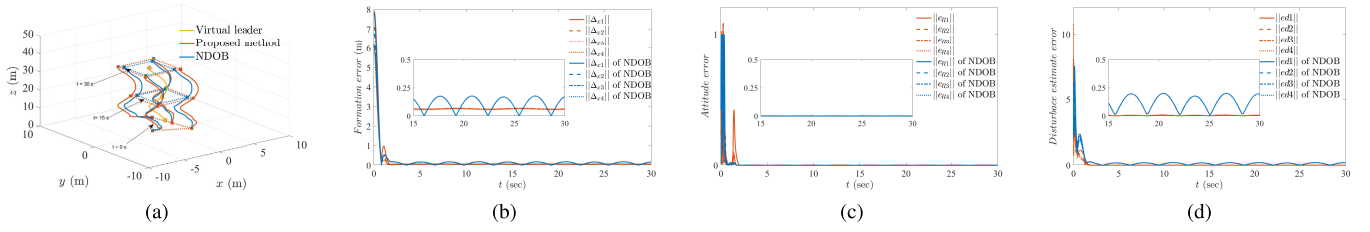


Fig. 3. The simulation results of the robustness comparison experiment are presented, where the orange line represents the proposed method and the blue line corresponds to the NDOB method. (a) Trajectories of the virtual leader and four follower quadrotor UAVs in 3D space, (b) Formation errors of each follower quadrotor. (c) Attitude error of each follower quadrotor. (d) The estimation error of the lumped disturbance, expressed as  $ed_i = \sqrt{\sum_{q=1}^N ((d_{fi,q} - \hat{d}_{fi,q})^2 + (d_{\tau i,q} - \hat{d}_{\tau i,q})^2 / N)}$ .

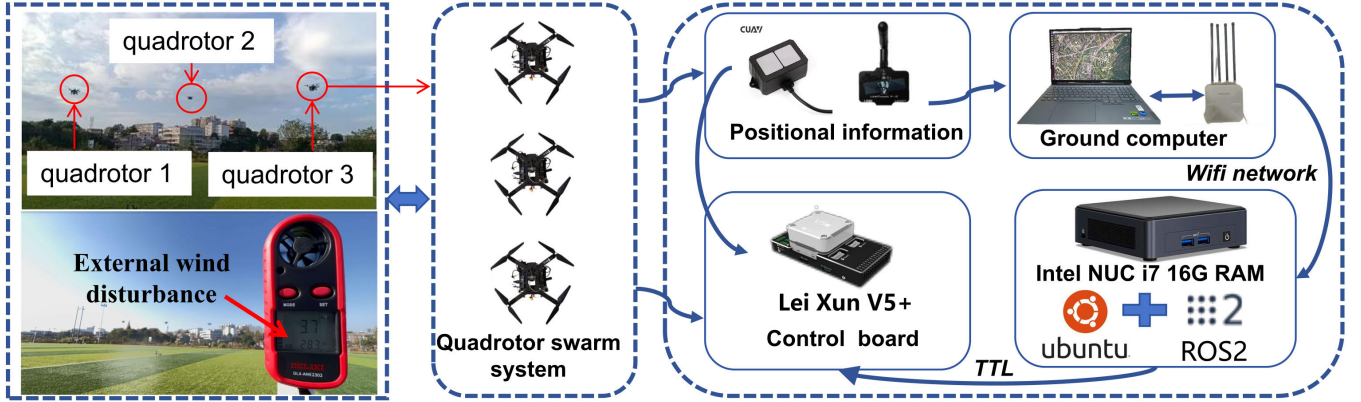


Fig. 4. Experimental setup for the QS system subjected to internal and external uncertainty.

### B. Robustness Comparison Simulation

The formation control of the nominal system is implemented with a geometric controller, and the disturbances are identified and compensated with a nonlinear disturbance observer (NDOB) from the literature [31]. The proposed method and the comparison method with NDOB were evaluated under identical simulation conditions to compare their respective anti-disturbance performance. Assuming that the system suffers from a small time-varying disturbance  $d_{fi} = [0.2 \sin(t) \ 0.2 \sin(t) \ 0.2 \sin(t)]^T$ ,  $d_{\tau i} = [0.3 \sin(t) \ 0.3 \sin(t) \ 0.3 \sin(t)]^T$ . The virtual leader's linear velocity is set as  $\dot{x}_0 = [-0.3 \sin(0.3t), 0.3 \cos(0.3t), 1]^T$ . Simulation results are presented in Fig. 3. Fig. 3(a) shows the trajectory of the QS system in 3D space, where the yellow line is the trajectory of the virtual leader, and the orange and blue lines are the trajectories of the four follower quadrotors in the proposed method and the NDOB method, respectively. The formation error and attitude error of the four follower quadrotors are illustrated in Fig. 3(b) and Fig. 3(c), respectively. Fig. 3(d) indicates the disturbance estimate error. The simulation results reveal that the NDOB method has a large error in the estimation of disturbances compared to the proposed method, which leads to a certain formation error. The comparative study shows that the proposed method has stronger robustness compared to the observer's method.

### C. Experimental Results

1) Experiment Setup: To demonstrate the practical effect of the proposed distributed robust formation controller, we conducted experimental validation in a real outdoor environment.

The experimental setup, as depicted in Fig. 4, comprises three quadcopter UAVs, a ground station, and a localization system. The localization system consists of UWB, barometer, and radar altimeter. Initially, four UWBs were deployed as base stations on the test site to establish an absolute coordinate system. Each quadrotor is equipped with a UWB module for determining the horizontal position information of the UAV. Subsequently, the altitude of the UAV is determined by integrating data from both the barometer and radar altimeter. The ground station computer and the onboard computer establish communication via a wireless network, while the onboard computer communicates with the autopilot PX4 through Micro DDS. Within the same local area network, each quadrotor in the unmanned swarm is assigned an identical topic name in ROS2, and communication between the quadrotors is facilitated through the utilization of ROS2's communication mechanism. All programs run on the robot operation system 2 (ROS2) foxy on the Ubuntu 20.04 system for distributed robust formation control. The mass of the UAVs is about 2.42 kg, the mass of each UAV will be inconsistent due to the inconsistency of battery models, the difference in mass is regarded as a model parameter uncertainty term, and the UAVs suffer from time-varying wind disturbances from the outside world during the experiments. The wind speed is between 0.0m/s and 3.8m/s.

2) Performance Evaluation: Evaluating the trajectory tracking performance of distributed robust formation controllers in the presence of UAVs with inconsistent mass subject to external time-varying disturbances. The experiments are set up with two different trajectories.

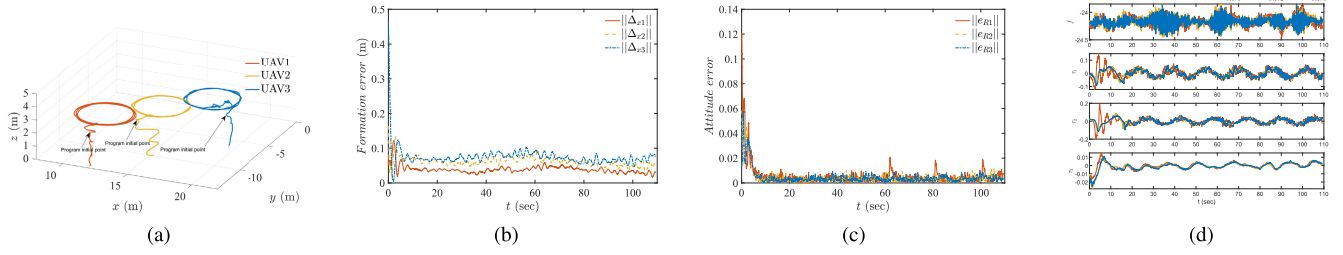


Fig. 5. Experiment 1: Circular trajectory formation control. (a) Tracking trajectories of three quadrotors in 3-D space. (b) Position errors in formation control. (c) Attitude errors in formation control. (d) The time evolution of net thrust force and torque inputs.

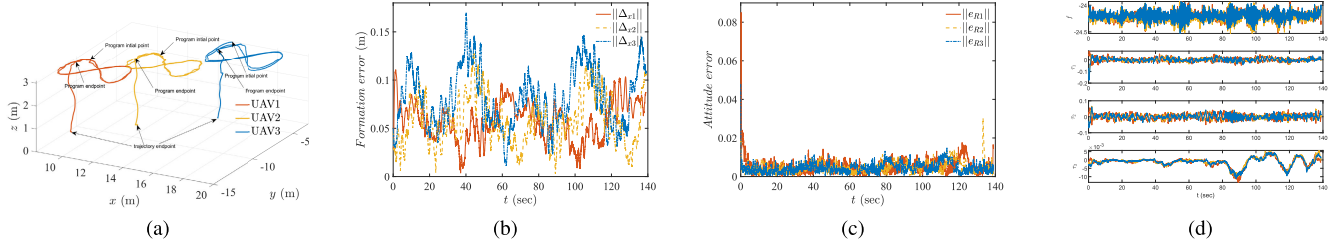


Fig. 6. Experiment 2: 8-shape trajectory formation control. (a) 8-shape trajectories in 3-D space. (b) Position errors in formation control. (c) Attitude errors in formation control. (d) The time evolution of net thrust force and torque inputs.

*Case I: Trajectory tracking formation with circle shape:* Set the initial position of each UAV as  $x_0 = [15, -10, 0]^T$ ,  $x_1 = [12.12, -13.77, 0]^T$ ,  $x_2 = [15.4, -10.22, 0]^T$ ,  $x_3 = [19.49, -6.31, 0]^T$ . The relative position of each follower UAV concerning the virtual leader is set as  $p_1 = [-3, -3, 0]^T$ ,  $p_2 = [0, 0, 0]^T$ ,  $p_3 = [3, 3, 0]^T$ . The reference trajectory of the virtual leader is set to be  $x_0 = [2 \cos(0.2t), -2 \sin(0.2t), 3]^T$ , the location of the center of the circle is  $[15, -8, 3]^T$ . Desired yaw angle is  $\psi_d = 0$ . The UWB, Radar Altimeter, and IMU are used to get position and attitude information at 50Hz. The experimental time is about 110 seconds. The experimental results are shown in Fig. 5. Fig. 5(a) is the 3D trajectory map, where the red, yellow, and blue lines are the tracking trajectories of UAV1, UAV2, and UAV3 respectively. The arrow in the picture indicates the position of switching program. The trajectory below the arrow is the trajectory of the UAV to be remotely controlled by the controller. Fig. 5(b) and Fig. 5(c) illustrate the experimental formation error and attitude error, respectively. The net thrust force  $f_i$  and torque inputs  $\tau_i$  over time are depicted in Fig. 5(d). The experimental results show that the distributed robust formation controller based on  $\mathcal{L}_1$  adaptation proposed in this paper can resist the effects of inconsistent mass of each UAV and external time-varying disturbances and quickly form the desired formation pattern.

*Case II: Trajectory tracking formation with 8 shape:* For case II we choose the more complex figure of eight as the tracking trajectory for each quadrotor follower. Set the follower UAV2 to have the same trajectory as the virtual leader UAV. The relative position of each follower UAV concerning the virtual leader is set as  $p_1 = [-3, -2.65, 0]^T$ ,  $p_2 = [0, 0, 0]^T$ ,  $p_3 = [3.5, 3.5, 0]^T$ . The reference trajectory of the virtual leader is set to be  $x_0 = [2 \sin(0.2t), -3 \sin(0.2t), \cos(0.2t), 3]^T$ , the location of the

center of the eights figure is  $[13.3, -8.4, 3]^T$ . Desired yaw angle is  $\psi_d = 0$ . The rest of the setup is the same as in case I. The experimental results are shown in Fig. 6. In Fig. 6(a), the red, yellow, and blue lines of 8 shape are the trajectory diagrams of UAV1, UAV2, and UAV3, respectively, in 3-dimensional space. The arrows in the figure point out the program start, and end and the trajectory end position. Fig. 6(b) and Fig. 6(c) present the formation error and attitude error, respectively. The net thrust force  $f_i$  and the torque inputs  $\tau_i$  with respect to time are illustrated in Fig. 6(d). As can be seen from the figure, the UAVs can form the desired formation and achieve complex formation trajectory tracking control in the presence of internal parameter uncertainty (mass inconsistency) and external time-varying disturbances (wind disturbances). Experiments demonstrate that the distributed robust formation controller based on the  $\mathcal{L}_1$  adaptation proposed in this paper has good formation control performance in complex situations. A video of the experiment can be found online at <https://youtu.be/q3RjaY3PIjY>.

## V. CONCLUSION

In this paper, a quadrotor-distributed formation control scheme without velocity feedback that can resist mass changes and external disturbances was proposed to achieve safe flight control of a QS system. The QS formation control was designed under a more communication resource-saving directed communication topology. Geometric controllers were designed in the system to achieve formation control of a QS in the ideal case. They accurately estimate unmeasurable velocity and higher-order derivatives of intermediate control variables for quadrotors using differentiators. The  $\mathcal{L}_1$  adaptive controllers were designed separately for the position and attitude loop subsystems to not only estimate the intrinsic uncertainty of the system but also to address the effects of external time-varying matching and non-matching disturbances. The

Lyapunov stability theory was employed to verify the stability of the formation controller, while simulation and experiments were conducted to demonstrate the effectiveness and favorable performance of the proposed method.

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